

SHAURYA SHARTHAK

Cybersecurity Engineer

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PROFESSIONAL SUMMARY

Security-minded engineer with hands-on **penetration testing** experience for MNC web applications and deep expertise in **AI/ML security**. Performed vulnerability assessments securing financial systems processing **2M+ transactions**. Built AI-driven security research tools and **red-teaming datasets** for adversarial AI testing. Strong Python skills with experience in **web scraping**, **automation**, and **API security**. Unique combination of offensive security skills and AI engineering — enabling next-generation **AI-powered threat detection** and **LLM safety research**.

TECHNICAL SKILLS

Security: Burp Suite, OWASP Top 10, SQL injection, XSS, CSRF, BOLA, IDOR, Broken Access Control, API Security

Offensive Tools: Penetration Testing, Vulnerability Assessment, Web Application Security, Red-Teaming

AI Security: Adversarial AI Testing, LLM Red-Teaming, Alignment Research, Uncensored Model Analysis

Languages: Python (Expert), SQL, Bash, C++

Infrastructure: Git/GitHub, Docker, Jenkins CI/CD, AWS (EC2/S3), Linux, REST APIs

AI/ML: PyTorch, HuggingFace, RLHF/GRPO, LangChain, Web Scraping (BeautifulSoup/Selenium)

EXPERIENCE

AI Engineer (Intern)

08/2024 – 01/2026

ResoluteAI Software Inc.

Bangalore, India

- **Secure System Design:** Architected hybrid retrieval pipeline (TF-IDF + BM25 + Neo4j + Redis) with secure API endpoints and access controls, improving hit-rate by **45%**.
- **Infrastructure Hardening:** Refactored inference pipelines with security-conscious design, achieving **Rs.25.8 Lakhs/year** cost savings and **30% latency reduction**.
- **Automation & Testing:** Agile sprints, Git feature branching, Postman API security testing, Jenkins CI/CD.

KEY PROJECTS

Penetration Testing — MNC Web Applications

2024

- Led deep-dive vulnerability assessments for **3+ MNC web applications** using **Burp Suite**, identifying critical flaws in authentication modules, data handling, and API endpoints.
- Discovered **SQL injection**, **XSS**, **CSRF**, **BOLA**, **IDOR** and **Broken Access Control** vulnerabilities in production systems.
- Improved security posture for high-traffic financial systems processing **2M+ transactions**, mitigating risks of data breaches and unauthorized access.
- Delivered actionable remediation reports with CVSS scoring and prioritized fix recommendations.

Mouse-OS — AI-Generated Security Research Tool

2024

- Developed AI-generated Remote Access Trojan (RAT) for **security research and red-teaming** purposes in Python.
- Demonstrated AI-assisted offensive security capabilities for threat modeling and defense development.

AI Red-Teaming & Adversarial Research

2024 – 2025

- Created **Toxic_qa_chatml** (6.9K rows) — adversarial QA dataset for LLM red-teaming and safety testing.
- Built **Unaligned-Thinking-o1** (159 rows) — unconstrained reasoning dataset exploring LLM behavior without alignment guardrails.
- Trained **Unaligned-Thinker-PHI-4** (15B, 15 likes) and **Unaligned-Thinker-Llama3.1-8B** — models for studying alignment boundaries and safety gaps.
- Contributed to RLHF annotation and red-teaming workflows at OpenTrain AI.

Automation & Scraping Tools | GitHub

2022 – 2025

- Built **AnimeDownloader** (5 stars, 6 forks), **IRCTC-Reservation-Finder**, **Microsoft-Rewards-Automator**, **Spotify-dl** — Python automation tools with web scraping and session handling.

PyTorch/XLA TPU Training Infrastructure | GitHub

2024

- Engineered distributed training framework for **7B–14B** parameter models on TPU; **35% HBM reduction**.

PUBLICATIONS

Tokenizer Flexibility in LLMs via Heuristic Adaptation and Supertoken Learning

arXiv:2505.09738 | 2025

- **Lead Author | 4 citations** — Model-agnostic framework with **2×** perplexity improvement; demonstrates rigorous security-conscious experimental methodology.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University

2022 – 2026

B.Tech (Hons.) Computer Science (AI/ML)

Bhilai, India

Jawahar Navodaya Vidyalaya — Dakshana Foundation Scholar (NDST 2018)

2018 – 2021

Senior Secondary (GPA: 3.86/4.0) — Secondary (GPA: 3.6/4.0)

Kurukshetra, India